PATENT APPLICATION DISCLOSURE

International Patent Cooperation Treaty (PCT) Filing Specification

Project Name:	Project Cool Power Core
System Title:	Automated Closed-Loop Atmospheric Remediation Engine & Multi-Frontier Micro-Purification Architecture
Primary Inventor:	Thomas G Coleman
Classification Fields:	USPC 60/517 (Free-piston engines); IPC F02B 71/04 (Linear alternators); CPC B01D 53/32 (Chemical gas purification by discharge); F01D 1/36 (Tesla blade-less turbines)
Document Status:	Official IP Legal Specification Matrix & Filing Disclosure Sheet

Section I: Independent System Claims

Claim 1. An autonomous, self-powered atmospheric remediation and multi-frontier micro-purification apparatus comprising:

- an intake manifold housing an acoustic cavitation matrix configured to transform an incoming fluid stream comprising ambient humidity, air, and volatile organic compound (VOC) particulate matter into a preheated cold steam aerosol;
- a constant-volume core combustion chamber lined with a Yttria-Stabilized Zirconia (YSZ) thermal barrier sleeve containing a dual opposed crankshaft-less piston assembly;
- a high-voltage plasma generation module driving an 8-stage Cockcroft-Walton voltage acceleration ladder configured to deliver a transient 33.6 kV nanosecond corona discharge across a zero-voltage switching (ZVS) resonant field into said core combustion chamber, structurally cracking said cold steam aerosol into an ionized radical matrix;
- a direct linear alternator assembly comprising N52H radial-magnetized magnet shafts integrated with said dual opposed piston assembly, configured to harvest the constant-volume expansion energy of said ionized radical matrix into a high/low split electrical bus network; and
- a downstream cyclonic condenser comprising a blade-less co-rotating Tesla disc turbine, configured to centrifugally separate liquid water fractions from processed sterile exhaust gases, wherein said liquid water fractions are fluidically re-routed back to said intake manifold without mechanical moving valves.

Claim 2. A method for reducing the chemical cracking activation threshold of an atmospheric working fluid within an autonomous remediation engine, the method comprising the steps of:

- routing a thermal feedback loop from a downstream exhaust block through a micro-channel copper heat exchanger embedded within a fluid storage reservoir;
- maintaining said fluid storage reservoir at a continuous preheated steady-state temperature of exactly 90°C, thereby inflating the internal fluid vapor pressure to 70.1 kPa to bypass approximately 83% of the liquid's latent heat of vaporization boundary; and
- injecting said preheated fluid directly into an intake air stream via a micro-metering cavitation nozzle array to establish a low-activation, radical-dense charge prior to plasma-chemical ignition.

Section II: Dependent Component Claims

Claim 3. The apparatus of Claim 1, wherein said intake fluid manifold is controlled by a non-electronic Graphene Fluidic Logic Gate Array, the topography comprising:

- a multi-tier monolithic interaction chamber completely coated in an atom-thick layer of chemical vapor deposition (CVD) graphene to establish an internal boundary-layer friction coefficient of ≤ 0.001;
- a bistable fluidic flip-flop amplifier configured to switch an internal high-velocity gas jet between distinct outlet channels using the Coandă effect driven by localized position-pressure ports; and
- a central fluidic AND-gate configuration configured to deliver a structural pneumatic pressure spike to trigger said ZVS plasma field exclusively when a piston position vector and an internal cylinder pressure vector converge simultaneously at a Top Dead Center (TDC) coordinate window.

Claim 4. The apparatus of Claim 1, wherein said dual opposed piston assembly is suspended friction-free within said YSZ thermal barrier sleeve via a concentric 5-micron aerostatic air bearing film, the routing lines comprising:

- a high-pressure feed gallery machined into the outer circumference of said YSZ sleeve, supplied with micro-filtered, dry exhaust bleed gas held at a continuous positive pressure profile of ≥ +250 kPa; and
- two distinct axial planes of sixteen circumferentially spaced, laser-drilled 0.12 mm diameter micro-metering orifices, configured to inject gas radially against said pistons to generate a geometric self-centering restoring force that scales inversely to the third power of the clearance gap (h^3).

Section III: Multi-Jurisdictional Scaling Claims

Claim 5. The apparatus of Claim 1, wherein the structural dimensioning features proportional geometric scaling constraints optimized for distinct ambient operational envelopes, comprising:

- a Micro-Scale Variant optimized for life-support suits, parameter-bound to a 10 mm bore × 15 mm stroke footprint to process a volumetric sweep of 2.5 m^3/hr of cabin air;
- a Medium-Scale Variant optimized for automotive cabin remediation, parameter-bound to a 25 mm bore × 40 mm stroke footprint to process a volumetric sweep of 50.0 m^3/hr of interior vehicle air; and
- a Macro-Scale Variant optimized for infrastructure smog scrubbing, parameter-bound to a 120 mm bore × 180 mm stroke footprint to process a volumetric sweep of 5,200 m^3/hr of municipal atmospheric gas masses.

CONFIDENTIAL INTELLECTUAL PROPERTY DISCLOSURE SHEET
DESIGNED BY THOMAS G COLEMAN — ALL RIGHTS RESERVED WORLDWIDE